

Ensemble benefit does not imply heterogeneity superiority

A Source ablation within the TG teacher

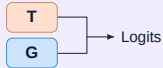
Grassmann-only
(G branch only)



Transformer-only
(T branch only)



Fused TG teacher
(T + G, $\alpha = 0.5$)



Positive transfer

+0.0410 NLL

Positive transfer

+0.1349 NLL

Positive transfer

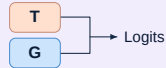
+0.1557 NLL

T-only: early clipping saturation;
interpret F-T gap cautiously.

Fused supervision transfers better than either component branch alone.

B Homogeneous control

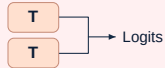
Heterogeneous TG ensemble
(T + G, $\alpha = 0.5$)



Positive transfer

+0.1557 NLL

Homogeneous TT ensemble
(T + T, $\alpha = 0.5$)



Positive transfer

+0.1681 NLL

TT yields lower student NLL than TG for all three matched students.

TT and TG teachers are not strictly quality- or training-matched.

Ensemble supervision helps, but neither architectural heterogeneity nor larger branch disagreement guarantees better transfer.